

Xuan Luo

6582 Calle Koral, Goleta, CA 93117 | xuan_luo@ucsb.edu | <https://luoxuan-cs.github.io/>

Research Focus

My research primarily focuses on Efficient Large Language Models:

- Computationally efficient inference in large language models (FlexiDepth, DiffSkip).
- Direct decoding of multiple tokens without relying on speculative decoding (DMTD).
- Context-adaptive efficient attention (AHA).

Education

University of California Santa Barbara, CA, USA

Sep 2023 – ongoing

Ph.D. Student, Computer Science

Advisor: Prof. Xifeng Yan

Xi'an Jiaotong University, Shaanxi, China

Aug 2019 – June 2023

B.E. Computer Science (Honor Program)

Preprints

Learning When Not to Attend Globally

2026

Xuan Luo, Jiaming Shan, Wesley Truong, Kailai Zhang, Xifeng Yan

Under review; positive feedback

[Preprint]

Token-Adaptive Representation for Attention

2026

Zhiyuan Wang, Xuan Luo, Xifeng Yan

Under review; positive feedback

Dual Dimensionality for Local and Global Attention

2026

Zhiyuan Wang, Xuan Luo, Sirui Zeng, Xifeng Yan

Preprint

[Preprint]

Publications

Direct Multi-Token Decoding

2026

Xuan Luo, Weizhi Wang, Xifeng Yan

COLM 2026

[PDF] [CODE] [MODEL] [WEBSITE]

Adaptive Layer-skipping in Pre-trained LLMs

2025

Xuan Luo, Weizhi Wang, Xifeng Yan

COLM 2025

Oral Presentation (top 24/418). Also invited to give a talk at NVIDIA Research

[PDF] [CODE] [MODEL] [DATASET] [WEBSITE]

DiffSkip: Differential Layer Skipping in Large Language Models

2025

Xuan Luo, Weizhi Wang, Xifeng Yan

ACL Findings 2025

[PDF] [CODE]

- LFSRM: Few-Shot Diagram-Sentence Matching via Local-Feedback Self-Regulating Memory** 2025
Lingling Zhang, Wenjun Wu, Jun Liu, Xiaojun Chang, Xin Hu, Xuan Luo, Yaqiang Wu, Qinghua Zheng
TPAMI 2025
[PDF]
- Bot or Human? Detecting ChatGPT Imposters with A Single Question** 2024
Hong Wang, Xuan Luo, Weizhi Wang, Xifeng Yan
COLM 2024
[PDF] [CODE] [DATASET]
- Progressive Attentional Manifold Alignment for Arbitrary Style Transfer** 2022
Xuan Luo, Zhen Han, Linkang Yang
ACCV 2022
[PDF] [CODE]
- Computer Science Diagram Understanding with Topology Parsing** 2022
Shaowei Wang, Lingling Zhang, Xuan Luo, Yi Yang, Xin Hu, Tao Qin, Jun Liu
TKDD 2022
[PDF] [CODE]

Professional Services

Reviewer: COLM 2026, COLM 2025, Neurips 2024, ACCV 2022

Experience & Projects

- TikTok**, San Jose, CA Jun 2026 – Sep 2026
Research Scientist Intern
TikTok Recommendation-LLMs, RL, GenAI
Mentor: Yang Zhang
- Large-scale pre-training for Recommendation-LLMs, including training acceleration and MoE model optimization.
- BioPACIFIC MIP (NSF Materials Innovation Platform)**, UCSB Apr 2025 – Jun 2026
Lead Developer / AI Researcher
- Full-stack developed a domain-specific Agentic AI system for scientific literature analysis and experimental result synthesis. [System Wiki]
 - Optimized the performance of the agent system and supported its production deployment.

Technical Skills

LLM Training: Pre-training, Supervised Fine-Tuning, Parameter Efficient Training.

Model Optimization: LLM Deployment and Optimization, Flex Attention, quantization, Triton, etc.

Agent & Applications: Agentic AI System Deployment and Performance Optimization.

Basics: PyTorch, Transformers, vLLM, Python, C++ , Java.